Inverse Trigonometric Function: Formula Sheet

Domain and range

$\sin^{-1} x$:

$$[-1,1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$\cos^{-1} x$: $[-1,1] \to [0,\pi]$

$\tan^{-1} x$:

$$R \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$\cot^{-1} x$: $R \to (0,\pi)$

$\sec^{-1} x$:

$$|x| \geq 1 \to [0,\pi] - \left\{\frac{\pi}{2}\right\}$$

$\operatorname{cosec}^{-1} x$:

$$|x| \geq 1 \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$$

Derivatives

$\sin^{-1} x$: $\frac{1}{\sqrt{1-x^2}}$; $\cos^{-1} x$: $-\frac{1}{\sqrt{1-x^2}}$

$\tan^{-1} x$: $\frac{1}{1+x^2}$; $\cot^{-1} x$: $-\frac{1}{1+x^2}$

$\sec^{-1} x$: $\frac{1}{|x|\sqrt{x^2-1}}$; $\operatorname{cosec}^{-1} x$: $-\frac{1}{|x|\sqrt{x^2-1}}$

Inequalities

$\sin^{-1}$:

$$\sin^{-1} x > \sin^{-1} y \iff 1 \geq x > y \geq -1$$

$\cos^{-1}$:

$$\cos^{-1} x > \cos^{-1} y \iff -1 \leq x < y \leq 1$$

$\tan^{-1}$, $\cot^{-1}$: $x > y$, resp. $x < y$

Property 1 and 2

Property 1: $\sin(\sin^{-1} x) = x$, $|x| \leq 1$; $\tan(\tan^{-1} x) = x$, $x \in R$

$\sin^{-1}(\sin x)$: $x - n\pi$ (n even), $n\pi - x$ (n odd)

$\cos^{-1}(\cos x)$: $|x - 2k\pi|$

$\tan^{-1}(\tan x)$: $x - n\pi$

$\cot^{-1}(\cot x)$:

$$x - \left[\frac{x}{\pi}\right]\pi$$

Periods: $\sin^{-1}(\sin x)$, $\cos^{-1}(\cos x)$: 2π; $\tan^{-1}(\tan x)$, $\cot^{-1}(\cot x)$: π

Properties 3, 4, 5

Odd: $\sin^{-1}(-x) = -\sin^{-1} x$, $\tan^{-1}(-x) = -\tan^{-1} x$

$\pi-$: $\cos^{-1}(-x) = \pi - \cos^{-1} x$, $\cot^{-1}(-x) = \pi - \cot^{-1} x$

Sum $\frac{\pi}{2}$: $\sin^{-1} x + \cos^{-1} x$, $\tan^{-1} x + \cot^{-1} x$, $\sec^{-1} x + \operatorname{cosec}^{-1} x$

Reciprocal: $\operatorname{cosec}^{-1} x = \sin^{-1} \frac{1}{x}$, $\sec^{-1} x = \cos^{-1} \frac{1}{x}$

$\cot^{-1} x$: $\tan^{-1} \frac{1}{x}$ $(x > 0)$, $\pi + \tan^{-1} \frac{1}{x}$ $(x < 0)$

Range:

$$(\sin^{-1} x)^2 + (\cos^{-1} x)^2 \in \left[\frac{\pi^2}{8}, \frac{5\pi^2}{4}\right]$$

Conversions

$x \geq 0$:

$$\tan^{-1} x = \sin^{-1} \frac{x}{\sqrt{1+x^2}} = \cos^{-1} \frac{1}{\sqrt{1+x^2}}$$

$0 \leq x \leq 1$:

$$\sin^{-1} x = \cos^{-1} \sqrt{1-x^2} = \tan^{-1} \frac{x}{\sqrt{1-x^2}}$$

Substitution results

$|x| \leq 1$:

$$\sin^{-1} \frac{2x}{1+x^2} = 2\tan^{-1} x$$

$x \geq 0$:

$$\cos^{-1} \frac{1-x^2}{1+x^2} = 2\tan^{-1} x$$

$|x| < 1$:

$$\tan^{-1} \frac{2x}{1-x^2} = 2\tan^{-1} x$$

$|x| \leq \frac{1}{2}$: $\sin^{-1}(3x - 4x^3) = 3\sin^{-1} x$

$\frac{1}{2} \leq x \leq 1$: $\cos^{-1}(4x^3 - 3x) = 3\cos^{-1} x$

$|x| < \frac{1}{\sqrt{3}}$:

$$\tan^{-1} \frac{3x - x^3}{1 - 3x^2} = 3\tan^{-1} x$$

$0 \leq x \leq 1$: $\cos^{-1}(2x^2 - 1) = 2\cos^{-1} x$

Addition and subtraction

$xy < 1$:

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{x+y}{1-xy}$$

$x, y > 0$, $xy > 1$:

$$\tan^{-1} x + \tan^{-1} y = \pi + \tan^{-1} \frac{x+y}{1-xy}$$

$xy > -1$:

$$\tan^{-1} x - \tan^{-1} y = \tan^{-1} \frac{x-y}{1+xy}$$

$$\sin^{-1} x + \sin^{-1} y = \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right)$$

$$\cos^{-1} x - \cos^{-1} y = \cos^{-1}\left(xy + \sqrt{1-x^2}\sqrt{1-y^2}\right)$$

Three angles:

$$\tan(A+B+C) = \frac{S_1 - S_3}{1 - S_2}$$

Half:

$$\tan\left(\frac{1}{2}\cos^{-1} x\right) = \sqrt{\frac{1-x}{1+x}}$$

Series

Split:

$$\tan^{-1}\frac{b-a}{1+ab} = \tan^{-1} b - \tan^{-1} a$$

$$\sum_{r=1}^{n} \tan^{-1}\frac{1}{1+r(r+1)} = \tan^{-1}(n+1) - \frac{\pi}{4}$$

$$\sum_{r=1}^{n} \tan^{-1}\frac{1}{x^2+(2r-1)x+r^2-r+1} = \tan^{-1}(x+n) - \tan^{-1} x$$